Z - TRANSFORMACIJE

$u[n] = u \cdot z^n$

$y[n] = H(z) \cdot u z^n$

$\downarrow$

$$H(z) = \sum_{n=-\infty}^{\infty} h[n] \cdot z^{-n}$$

BILATERALNA
ILI
DVOSTRANA Z. TRANSF

Primjer)

$$x[n] = \{ \underset{n=0}{1}, \underset{n=1}{2}, \underset{n=2}{3}, 3, 2, 1 \}$$

$$X[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} = 1 \cdot z^{-0} + 2 \cdot z^{-1} + 3 \cdot z^{-2} + 3 \cdot z^{-3} + 2 \cdot z^{-4} + 1 \cdot z^{-5}$$

$$= 1 + 2 \cdot \frac{1}{z} + 3 \frac{1}{z^2} + 3 \cdot \frac{1}{z^3} + 2 \cdot \frac{1}{z^4} + \frac{1}{z^5}$$

$$X[z] = \frac{z^5 + 2z^4 + 3z^3 + 3z^2 + 2z + 1}{z^5}, \quad z \in \mathbb{C}, \; z \neq 0 \; !$$

* ideja z-transf ↑

Primjer: $x[n] = \alpha^n \mu[n] = \{ \dots 0, 0, \underline{1}, \alpha^2, \alpha^3, \alpha^4, \dots \}$

$\alpha \in \mathbb{C}$

$$X[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} = \sum_{n=-\infty}^{\infty} \alpha^n \cdot \mu[n] z^{-n} = \sum_{n=0}^{\infty} \alpha^n \cdot z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

↳ geom red

da bi sumu uopće mogli izračunat, član $|\alpha z^{-1}| < 1$

$$X[z] = \frac{1}{1 - \alpha z^{-1}} = \frac{z}{z - \alpha}$$

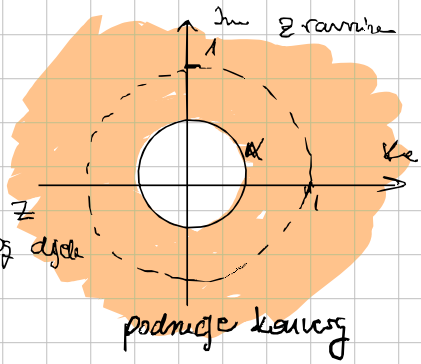
$$\left| \frac{\alpha}{z} \right| < 1$$

$|\alpha| < |z|$

npr. za $\alpha = 0.75$

$$X[z] = \frac{z}{z - 0.75} \longrightarrow \boxed{|z| > 0.75}$$

⇒ treba nam z iz otvorenog dijela



Primjer: $x[n] = -\alpha^n \mu[-n-1]$ → počinje u $-\infty$, završava u -1
(-1 je zadnji trenutak kad step postoji)

$$X[z] = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} = \sum_{n=-\infty}^{\infty} -\alpha^n \mu[-n-1] z^{-n} = -\sum_{n=-\infty}^{-1} \alpha^n z^{-n} =$$

→ izraz kao prije ali drugačije granice? hmm....

promjeni granice

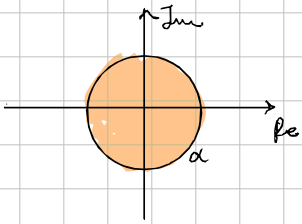
$$= -\sum_{n=+1}^{\infty} \alpha^{-n} \cdot z^{n} = -\sum_{n=0}^{\infty} \left(\frac{z}{\alpha}\right)^n + \alpha^{-0} \cdot z^0 = -\frac{1}{1-\frac{z}{\alpha}} + 1$$

dodali smo 1 element pa ga moramo oduzeti!

$$X[z] = \frac{z}{z-\alpha}$$ ⟹ ISTI IZRAZ a ovaj je ANTIKAUZALAN, a prije je bio KAUZALAN!

$\left|\frac{z}{\alpha}\right| < 1$

$|z| < 1$



→ ali ovo je područje konvergencije za ovaj slučaj!

Redefinicija → odsad pričamo samo o KAUZALNIM signalima
⟹ JEDNOSTRANE z-transf.

- ne moramo pisati i crtati područje konv.

$$X[z] = \sum_{n=0}^{\infty} h[n] \cdot z^{-n}$$

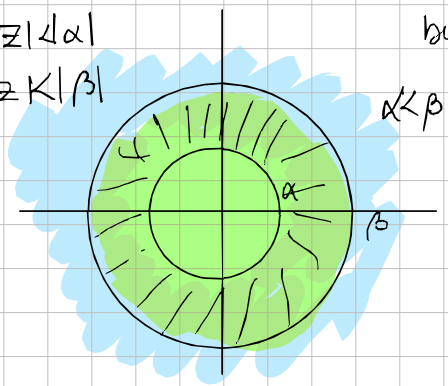
* Zadatak prije → da npr zbrojimo dva rj. otprije

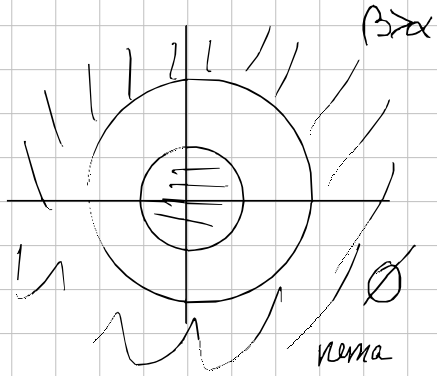
$$x[n] = \alpha^n \mu[n] + \beta^n \mu[-n-1] \longrightarrow X[z] = \frac{z}{z-\alpha} - \frac{z}{z-\beta}$$

budući da znamo transformacije

$|z| < |\alpha|$

$|z| < |\beta|$





nema preklapanja

→ z-transf. možemo naći samo ako je $\alpha < \beta$